Safe Paths to School: Evidence from Bogotá's Walking to School Caravans*

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Abstract

In many low- and middle-income countries, most children walk to school through dense and often unsafe neighborhoods. These commutes create daily coordination costs and safety concerns for parents, which undermine attendance. Despite its importance, commuting remains an overlooked dimension of education policy. This paper evaluates walking to school caravans (WSC), in which trained monitors accompany groups of children along safe routes. We study Bogotá's *Ciempiés* program, launched in 2019 and rolled out across public primary schools. Using administrative data from 2016-2024 and a differences-in-differences design that exploits the staggered adoption of the program, we estimate causal effects on educational outcomes and road safety. Overall, WSC reduce school switching by 6.8 percentage points and branch switching by 10.9 points (over 40% declines from pre-treatment rates for the control group). We also find suggestive reductions in grade repetition and dropout, though no effects on traffic accidents. At just USD 285 per child annually, WSC are a cost-effective, scalable intervention that eases family coordination burdens and improves student permanence, a model that can be replicated in other dense urban settings worldwide.

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1 Introduction

School attendance is a prerequisite for learning, yet many children in Low- and Middle-Income Countries (LMICs) continue to miss school regularly due to logistical and coordination barriers in school transportation. In dense urban settings, where most children walk to school, commutes themselves can become a daily hurdle. Long distances, dangerous or high-traffic neighborhoods, and the absence of formal transport systems force families into difficult trade-offs: parents can accompany their children to school but lose hours of work, send them alone but worry about safety, or keep them home when coordination fails. Irregular attendance undermines learning outcomes (Aucejo and Romano, 2016; Liu et al., 2021), raising dropout risk (Koppensteiner and Menezes, 2021), and affecting peers (Goodman, 2014; Bennett and Bergman, 2021). Despite its importance, commuting has received far less attention in education policy than classroom inputs or curricula.

This paper evaluates a large-scale intervention designed to directly reduce commuting and coordination costs: Walking to School Caravans (WSC). These programs recruit trained adults to accompany groups of children along predetermined safe routes, with designated meeting points where parents can drop off and pick up their children. Children walk together, the youngest hand by hand, learning about traffic norms and safety along the way. WSC thus lower the cost of attendance by reducing the time burden on parents, alleviating safety concerns, and providing children with additional exercise and road safety knowledge.

We study one of the largest implementations of WSC, Bogotá's *Ciempiés* (spanish for Centipede) program. Launched in 2019 and rolled out staggeredly across public primary schools, *Ciempiés* provides quasi-experimental variation in treatment timing. We combine rich administrative data from Bogotá's Secretaries of Education and Mobility with a difference-in-differences design to measure the program's impacts. We find that WSC can substantially improve children's educational trajectories. Overall, enrollment in *Ciempiés* reduces school switching by 6.8 percentage points and branch switching by 10.9 points. We also find a reduction of 5 percentage points in grade repetition, though this estimate is imprecisely estimated. These effects are large in magnitude and suggest that WSC strengthen children's attachment to their school. At a cost of only USD 285 per child per year, the program is highly cost-effective compared to traditional mentoring or school transportation interventions.

Despite these educational benefits for children, we find no evidence that WSC improved road safety. Using traffic accident data, we compare treated and not-yet-treated census tracts and find no impact on accidents involving children under 12. School-issued accident alerts show similar null effects, if anything pointing to small increases in incidents. These results likely reflect the counteracting forces of safer routes and traffic education, on the one hand, and reduced adult supervision (two monitors for up to 50 children instead of parents), on the other. The value of WSC thus lies less in reducing accidents than in easing the daily coordination burden of getting children to school. In contexts where small disruptions in parents' schedules or transportation systems translate into lost school days, interventions like WSC can stabilize attendance and keep children in school.

The case of Bogotá illustrates why commuting is such a central barrier to education. In 2024, two-thirds of all school trips in the city were made on foot, while only 16 percent were made by bus and 9 percent by public transport. For children in public schools, who come from poorer families and are less able to pay for private transportation arrangements, walking is more predominant. Among families who do purchase transportation, the financial burden is substantial: households report average monthly expenditures of COP 170,000 (roughly USD 43), 1.5 times the cost of *Ciempiés* on a per-child basis. This cost is particularly high when compared to Colombia's minimum wage, about USD 240 per month in 2023, meaning that formal transportation absorbs a large share of family income. For most low-income families, therefore, the only option is to accompany children by foot, rearranging work schedules and increasing the fragility of attendance.

Time burdens are also unevenly distributed across socioeconomic status. Data from Colombia's Quality of Life Survey (QLS) show that children from the lowest socioeconomic strata spend nearly 13 minutes walking to school on average, compared to under 10 minutes for children from higher strata. The difference may appear small, but multiplied across two daily trips and over the course of an academic year, it represents dozens of additional hours of commuting time borne disproportionately by the poorest families. In Bogotá, there's also a clear socioeconomic in school distance/travel time: children in extreme poverty (SISBEN score of 0) travel on average 2.2 kilometers to school, compared to 1.4 kilometers for children in less vulnerable households. Families in informal settlements often live farther from public schools and are more likely to be assigned to distant schools with available seats. These constraints mean that the children most at risk of dropout also face the most burdensome commutes.

Within this context, Ciempiés provides a community-based solution that reduces coordination costs and eases the burden of commuting with children for parents. By organizing children into supervised walking groups with designated meeting points, the program allows parents to delegate responsibility for part or all of the commute. This is particularly valuable given that the average route in the program covers 1.5 kilometers and requires about 40 minutes of walking time. For a working parent, this time commitment can make the difference between maintaining employment and having to forgo income. The program thus addresses not only safety concerns but also the opportunity costs of accompanying children, which are particularly high for parents in informal labor markets.

Traditional education policies have sought to increase attendance primarily through demand-side incentives such as Conditional Cash Transfers (CCT), which require 80–85 percent attendance and provide income support to poor families (Baird et al., 2014; Fiszbein et al., 2009; Millán et al., 2019). Other interventions reduce parental monitoring costs by providing information about attendance or assignments (Bergman, 2021; Berlinski et al., 2025; Lichand et al., 2024). Supply-side policies, such as school construction, reduce distances but are costly and slow to scale in urban areas (e.g., Burde and Linden, 2012; Dinerstein et al., 2023; Duflo, 2001; Khanna, 2022). Transport-based solutions represent a third approach, but their effectiveness is mixed. In high-income countries, busing has been widely used to increase racial integration, but recent evidence suggests it is not cost-effective: Angrist et al. (2022) find that busing in Boston and New York improves integration but has no effect on achievement, despite very high costs. In LMICs, bicycles have been shown to improve girls' enrollment, test scores, and empowerment (Muralidharan and Prakash, 2017; Fiala et al., 2025), but they are limited to children old enough to ride and to areas with safe roads. WSC, by contrast, are low-cost, fast to scale, and well-suited to the realities of dense cities where most children walk to school.

The mechanisms through which WSC operate are multiple. They ease the time and coordination costs for families, particularly those with inflexible work schedules or multiple children to care for. They alleviate safety concerns by ensuring children are accompanied by trained monitors and peers. They provide children with daily exercise and practical knowledge of traffic rules. And they create a social commuting experience that may strengthen peer ties and children's sense of belonging in school. These features distinguish WSC from other transport interventions, where the child is a passive rider rather than an active participant.

Our findings contribute to several strands of literature. We place commuting at the center of education debates, showing how overlooked coordination costs affect school attendance and permanence. We provide the first rigorous evaluation of WSC, complementing studies on bicycles in LMICs and buses in high-income countries (Angrist et al., 2022; Setren, 2024; Muralidharan and Prakash, 2017; Fiala et al., 2025). We add to research on urban safety and education, showing that programs designed to improve commuter safety for children may have limited effects when they substitute parental supervision (Bauernschuster and Rekers, 2022; Koppensteiner and Menezes, 2021). Finally, we contribute to the literature on neighborhood effects and place-based policies (Chetty et al., 2016; Chetty and Hendren, 2018a,b) by showing that intervening neighborhoods and creating safe paths to school can increase school attachment without requiring families to relocate away from their communities and social networks.

More broadly, our results highlight commuting as a critical but neglected margin of education policy. In cities where most children walk through dense and often dangerous neighborhoods, WSC represent a scalable model to improve student attendance and permanence at very low cost. Beyond easing parental coordination, they provide children with exercise and safety training and foster a stronger attachment to school. The lessons extend beyond LMICs: many developed countries face similar challenges in urban commuting, making WSC relevant as a complementary strategy to buses or other forms of school transport.

The rest of the paper proceeds as follows: Section 2 describes the setting and data. Section 3 outlines the differences-in-differences strategy, and Section 4 presents results. We conclude in Section 5.

2 Setting and Data

Our study focuses on Bogotá, the capital and largest city in Colombia, whose public school system served an estimated 1,142,000 preschool, primary, and secondary students in 2024. Families in Bogotá can choose between public and private schools, with public school admissions coordinated through a centralized assignment system. In 2024, 63% of students attended public schools, while 37% were enrolled in private institutions. The public school choice system transitioned from a Boston-style mechanism to a Deferred Acceptance algorithm during 2022–2023. Our analysis focuses exclusively on the public sector, as the *Ciempiés* program operates only in public schools.

Parents in Bogotá have numerous public school options, and while distance plays a role in school choice, families do not always choose the nearest public school. This behavior reflects the consideration of multiple factors—such as school quality, perceived safety, and convenience regarding caregivers' work.[1] Yet, this choice flexibility is not the same across socioeconomic groups, creating systematic disparities in school commuting patterns.

Figure 2 presents a binscatter regression that reveals a socioeconomic gradient in school distance. Children with lower SISBEN III scores, which indicate higher vulnerability, tend to live farther from their school. Those in extreme poverty (score = 0) travel, on average, 2.2

[1] Many families, especially those with young children, enroll their children in schools close to their workplaces to facilitate drop-off and pick-up.

kilometers to school, while children with higher (though still vulnerable) scores travel around 1.4 kilometers. This pattern suggests that the poorest families face additional barriers to access, potentially due to informal housing in areas with poor public provision of education or late assignment to schools with available slots but in distant locations.[2]

The gradient in school distance translates into concrete burdens for vulnerable families. Figure 5 demonstrates that average travel times to school vary systematically by socioeconomic strata among children who commute by foot or bike. Children from the lowest strata (1 and 2) spend significantly more time commuting than those from higher strata, with strata 1 children requiring nearly 13 minutes on average compared to under 10 minutes for strata 4 children. These differences impose higher time costs and additional coordination challenges on low-income households, making consistent school attendance more challenging to maintain.

The transportation choices available to families reflect both preferences and constraints. Figure 3 illustrates that walking remains the most common way to commute to school in Bogotá, accounting for 66.4% of all school trips. This pattern is more pronounced in public schools, where students are significantly less likely to use paid transportation than their counterparts in private schools. The data show that only 16.4% of students use school buses, 9.4% rely on public transport, 4.7% travel by private vehicle, and 3.0% use bicycles.

For families who do pay for school transportation, the financial burden is substantial. Figure 4 presents the distribution of monthly transportation expenditures among households that report paying for school transportation services. On average, the amount that families pay per month for transportation services is approximately COP 170,000 (roughly USD 43). This amount is nearly double the cost that the city pays per child per month in *Ciempiés* (USD 23.75), highlighting the program's cost-effectiveness relative to private transportation alternatives.

The pattern that emerges is that many public school students live relatively far from school but cannot afford paid transportation options. This creates a challenge for low-income families, who must either invest significant time accompanying children on long walks to school or find alternative arrangements that may compromise safety or reliability.

In sum, while Bogotá offers choice within its public education system, geographic, logistical, and financial frictions constrain that choice for the poorest families. The consequences of this are longer travel times, greater need for coordination, and limited use of formal trans-

[2] While in many LMICs richer families live farther from schools due to suburban housing patterns (Beuermann et al., 2023; Hofflinger et al., 2020), the inverse gradient in Bogotá reflects the concentration of urban poverty in poorly connected peripheries and the challenges of navigating the centralized assignment system.

portation services. This can lead to inconsistent school attendance and exacerbate inequality in access to education. The *Ciempiés* program operates within this context, attempting to address the coordination problem parents face in bringing their children to school every day by providing a community-based transportation solution.

2.1 The Ciempiés Program

Bogotá's Secretaries of Education and Mobility launched *Ciempiés* in 2019. The program was designed to address safety concerns and coordination challenges faced by families whose children walk to school, particularly in unsafe areas and with high incidence of traffic.

The program operates through supervised walking caravans that transport children to and from school along predetermined safe routes. In each participating school, program staff map out an optimal walking path that avoids crime hotspots, minimizes exposure to heavy traffic, and connects the school with the highest concentration of children who walk to school. Two trained monitors lead organized walking caravans of up to 50 children on each route.

The key innovation of *Ciempiés* lies in its meeting point system, illustrated in Figure 1. Along each safe path, the program establishes designated meeting points where parents or caregivers can drop off and pick up children. This system reduces the coordination burden on families in several ways. Parents no longer need to accompany their children the entire way to school. Children benefit from adult supervision and peer interaction during the commute. Families gain flexibility in their daily schedules since the meeting points are strategically located to be accessible from multiple residential areas.

The program targets kindergarten and primary school children (ages 5-10) attending public schools. While there is an eligibility criterion requiring students to live within 700 meters of a meeting point, this criterion is not strictly enforced in practice, allowing program staff to accommodate families based on local needs and logistical considerations.

Ciempiés expanded through a staggered rollout that provides the quasi-experimental variation for this study. Starting with a pilot in 2018, and a first phase of the rollout in 2019, the program has grown steadily, reaching 53 school shifts across 9 of Bogotá's 19 localities by 2024. Figure 6 shows the spatial distribution of the program as of 2022. The program is scheduled to expand to an additional 75 schools through 2025-2030.

The cost of the program is approximately USD 285 per child per year (at 2023 prices), which includes monitor salaries, program administration, and equipment. This cost is significantly lower than formal transportation alternatives or intensive educational interventions, making it an attractive option for resource-constrained educational systems.

2.2 Data

We combine administrative data from three institutions (two at the local and one at the national level). Our three data sources are the following:

Secretary of Mobility. From Bogotá's Secretary of Mobility, we obtain detailed information about the Walking to School Buses (WSB) operation, including application and enrollment records for all participating children with their geocoded home addresses. The dataset includes daily attendance records for the walking caravans and documents which parent or caregiver dropped off and picked up each child at the meeting points.

Additionally, the Secretary of Mobility provided the universe of traffic accident reports in Bogotá from 2015-2024. These records include precise geolocation data, accident severity, victim demographics (age, gender), mode of involvement (pedestrian, passenger, driver, cyclist, motorcyclist), and timestamps. This comprehensive dataset allow us to examine effects of the program on accidents involving children of primary school age, and spillover effects to other age groups and geographic areas.

Secretary of Education. From Bogotá's Secretary of Education, we access the SIMAT (Integrated System of Student Enrollment), an annual registry of all students enrolled in Colombian public schools from 2016-2024.[3] This dataset includes socioeconomic information such as ethnicity, disability status, country of origin, socioeconomic stratum[4], and SISBEN scores. The data allows us to look at outcomes like grade progression, school switching, and dropout.

We also have access to records of alternative mobility programs and transportation subsidies from 2022 to 2024. We merge these records with SIMAT data to obtain a comprehensive picture of the transportation assistance available to households. The other mobility programs are *Al Colegio en Bici*, a program that organizes caravans of children biking to school with bicycles lended by the city government, and *Biciparceros*, which organizes groups of children (older than 14) to go to school with their own bikes, providing supervision and mechanical assistance along the way. Figure A.3 shows the shares of children in the different mobility programs (including *Ciempiés* and transportation subsidies), by deciles of distance to school.

Our learning outcomes come from the *Plataforma de Apoyo Escolar* (2015-2024), a system that the Secretary of Education provides to schools for generating school transcripts.

[3]For private schools, we only have data from 2020 onwards, since before that year they were not in SIMAT.

[4]The socioeconomic stratum, known as *estrato* in Spanish, refers to Colombia's socioeconomic stratification system, which categorizes households based on their affluence using neighborhood and dwelling characteristics, with one corresponding to the poorest, and six to the richest households.

From these transcripts we retrieve the end-of-year math and language grades, and in some cases, teacher-recorded absences and measures of parental involvement. Teachers assign grades at their discretion within each school, but they must use a national scale with four achievement categories: low, basic, medium, and high.

Finally, we also have access to novel data on school issued alerts from the *Sistema de Alertas*, which schools have to report by law to the Secretary of Education. The data covers behavioral alerts, and violence and accident reports from 2014-2024, and includes the time and description of the event, and the identity of the aggressor, disruptive kid or victim involved. We use the alerts issued for any accident that happened on the way to school to complement our analysis on the impact of *Ciempiés* on road safety outcomes.

Instituto Colombiano para la Evaluación de la Educación (Icfes). From Icfes, we obtain standardized test scores from the *Pruebas Saber* 3, 5, and 9 for 2017. This was the last year that standardized tests were applied broadly to primary school students before an interruption that lasted until 2022. After that year, only a representative sample of schools is tested.[5] These data allow us to assess balance between schools that adopted *Ciempiés* at some point between 2019 and 2024 and future adopters (never-treated group) before the program began, providing an indirect way to evaluate parallel trends.

Table 1 summarizes the characteristics of children participating in *Ciempiés* from 2021-2024. Participants are typically 8 to 9 years old (corresponding to third and fourth grades). Half of them are girls. The average child in our sample comes from socioeconomic stratum 2, indicating these are vulnerable children from low-income families. The data also reveal a higher representation of Venezuelan children and victims of Colombia's armed conflict, particularly in recent years. These groups face additional barriers to accessing education. For comparison, 7% of Bogotá's public school students were Venezuelan, and 4% were victims of Colombia's armed conflict in 2024, versus 13% and 8% in our sample.

Table 2 provides descriptive statistics for the walking paths from 2020-2024. The average path length is approximately 1.5-1.6 kilometers, requiring 40-42 minutes of group walking time. The program serves different school shifts, with a relatively balanced distribution between morning and afternoon shifts, plus some full-day programs.

These path characteristics highlight several important features of the program. First, the walking distances are substantial, which underscores why parents may have safety con-

[5] The *Pruebas Saber* evaluate student competencies in math, language, natural sciences, and citizenship across 3rd, 5th, and 9th grades. The 2017 edition only administered language and math tests. Individual test scores are categorized into four performance levels: insufficient, minimum, satisfactory, and advanced. Table A.1 describes each of these categories.

cerns and used to accompany their children to school or arrange a transportation service in the absence of the program. Second, the 40+ minute (one way) walking time represents a considerable daily time investment for families, making the supervised caravan model particularly valuable for working parents who cannot accompany children for such extended periods.

3 Empirical Strategy

To estimate the causal impact of *Ciempiés* on children's educational outcomes and road safety, we exploit the program's staggered rollout across Bogotá's public schools between 2019 and 2024. We implement a Difference-in-Differences (DiD) approach using the methodology proposed by Callaway and Sant'Anna (2021), which allows us to estimate group-time Average Treatment Effect on the Treated (ATT) for cohorts of children that began participating in different years.

This method compares the evolution of outcomes for treated children in schools where *Ciempiés* started operating relative to children in schools that have not yet implemented the program, but where the program expanded later or is set to expand in 2025-2030. Specifically, for each treatment cohort g and year t, we estimate:

$$ATT(g,t) = \mathbb{E}[Y_{it} - Y_{ig-1} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{ig-1} \mid G_i = g'], \text{ with } g' > t \tag{1}$$

where Y_{it} is the outcome of interest (e.g., dropout, grade repetition, switching schools or school branches, math and language grades) for student i in year t, and $G_i = g$ indicates the year cohort g in which the student started treatment. We use as controls all students in schools that have not been treated yet by time t (so cohorts $g' > t$). For the earlier cohorts, these include not-yet treated, and never-treated students, while for the last treated cohort (the one that started with *Ciempiés* in 2024), these include only never-treated students ($g = \infty$). In Equation(1), we include age and gender as controls.

The main assumption we have to make for the internal validity of our estimates is the parallel trends assumption based on not-yet-treated units, which states that in the absence of treatment, average potential outcomes of the group first treated in g and the group not-yet-treated in $t > g$, would have evolved in parallel.

We compute event-study estimates by grouping treatment effects across different cohorts by the number of years that have passed since program implementation:

$$ATT_l^w = \sum_g w_g \cdot ATT(g, g+l) \quad (2)$$

where l indexes the number of years since treatment began, and w_g represents the relative size of cohort g. This aggregation enables us to examine dynamic treatment effects and assess whether the program's impact grows or fades over time.

In our main specifications, we focus on a set of educational outcomes at the individual level using SIMAT: school switching (including switching between branches or *sedes*), grade repetition, and dropout. We measure dropout as a dummy variable that takes the value of one if, by the end of school year t, the student i didn't enroll in the public school system the following year. We define grade repetition in t as being in the same grade as in $t-1$.

To estimate the unintended effects on traffic accidents, we apply a similar DiD framework to traffic accident data but at the census tract level. We compare the number of accidents in census tracts intersecting with operating school paths to those in census tracts that intersect not-yet-treated school paths.[6] Additionally, we expand the not-yet-treated control group to include a never-treated control group consisting of all untreated census tracts in the *localidades* where *Ciempiés* operates.

Since our data on accidents is anonymized, we can't establish whether the accident victim is attending any of the schools in our sample. As such, our estimates recover both the effect of the program and any spillover effects that may occur if other pedestrians or drivers change their behavior along the safe paths to school. For this reason, we complement our analysis by focusing on the school issued alerts on student road accidents, which schools need to report to the Secretary of Education. We estimate Equation 1 at the school level for all the periods before and after program implementation, and for all cohorts, and also present these results in the next section.

4 Results

Our empirical strategy provides the causal impact of the *Ciempiés* WSB program on children's educational outcomes and road safety. Before turning to our main results, we examine whether schools that adopted *Ciempiés* at some point between 2019-2024 differed systematically from the not-yet adopters where the program will expand in the future. Table 3

[6] We use a catchment area of 250m around school paths to define treated versus control census tracts. Figure 7 illustrates the procedure that we follow.

reports means and standardized differences of baseline covariates measured in 2017-2018 (before any school adopted the program), including demographics, socioeconomic strata, traffic accidents on the way to school, and standardized test scores from the 2017 Saber assessments. We present both unweighted and student-weighted normalized differences.

The balance in covariates between adopters and non-adopters provides an indirect way of assessing the plausibility of the parallel trends assumption (Baker et al., 2025). If treated and control units are similar in observable variables that explain potential education outcomes and trends, this strengthens the claim that potential outcomes would have evolved in parallel in the absence of treatment. Our results in Table 3 show relatively small standardized differences between adopters and non-adopters, with most falling below 0.25 in absolute value—a commonly used threshold for meaningful imbalance (Baker et al., 2025). This evidence supports the credibility of our identification strategy and complements the graphical inspection of pre-trends that we discuss below.

4.1 School Switching

We now turn to the main results from our DiD estimation using the Callaway and Sant'Anna (2021) methodology, exploiting the staggered rollout of the program across Bogotá's public schools between 2021 and 2024.

Figure 8 presents our dynamic estimates of the impact of *Ciempiés* on children's educational trajectories. The corresponding estimates are reported in Table 4.

The event-study plots in Figure 8 show the dynamic ATT for four key outcomes: dropout, grade repetition, school switching, and switching between school branches. Each panel displays treatment effects relative to the year before program implementation, aggregated among the different cohorts, with negative time periods indicating pre-treatment periods and positive values representing post-treatment effects. Since the 2021 cohort is the only one we can observe for three years after treatment, and it is the smallest (N= 771), our results for $t = 3$ (and $t = 2$ for dropout) are noisy.

Panel C of Figure 8 shows that Ciempiés significantly reduces school switching. Overall, *Ciempiés* decreases school switching by 6.8 percentage points. Relative to the pre-treatment school switching rate of 16.5% in the control group, these represent a reduction of 41%. We find even more pronounced effects for the switching between school branches variable. Figure 8, Panel D shows that *Ciempiés* reduces branch switching by 10.9 percentage points. Compared to the pre-treatment branch switching rate of 25.8% in the control group, these effects represent a reduction of 42%. These large effects suggest that the program substantially increases children's attachment to their school by reducing transportation coordination challenges, which often force families to change their children's schools from one year to the next.

The pre-treatment coefficients in both panels show no systematic differences between treatment and control groups before program implementation, supporting the parallel trends assumption underlying our identification strategy. The growing magnitude of effects over time suggests that the benefits of the program compound as families become more accustomed to the *Ciempiés* routine and the social bonds within the walking caravans strengthen.

4.2 Grade Progression and Dropout

For grade repetition and dropout, shown in Figure 8, Panels A and B, respectively, we observe point estimates suggesting program benefits, though these effects are imprecisely estimated with large confidence intervals. Panel B shows that *Ciempiés* reduces grade repetition by 5 percentage points. Compared to the pre-treatment grade repetition rate of 2.3% in the control group, this reduction is substantial, though the statistical imprecision warrants caution in interpretation. Panel A suggests the program reduced dropout by 1.3 percentage points in the year it was adopted (t=0), representing a 13% reduction from the pre-treatment dropout rate of 9.75% in the control group. Although this effect is statistically significant, the coefficients for t=1 and t=2 jump around zero, resulting in an overall effect of 1.4 percentage points that is not statistically different from zero. These results reflect our limited power for long-term outcomes.

4.3 Results on Accidents

While the program was designed with road safety as a key component, our analysis of traffic accident data reveals more nuanced results. Figure 9 presents event-study estimates for accidents involving children 12 years old or younger in census tracts overlapping with the *Ciempiés* school paths.

Contrary to the program's safety objectives, we find no significant impact on traffic accidents involving primary school-age children. The event-study coefficients fluctuate around zero throughout both pre- and post-treatment periods, with confidence intervals that include zero for all time periods. This null effect persists whether we use only not-yet-treated units as controls or add never-treated units to the comparison group.

Our null results on accidents do not necessarily indicate that the program lacks safety value for children. Since we can't observe in our accident data whether the kid involved in the accident is enrolled in any of our treated schools, it is still plausible that the program may be increasing road safety for children in *Ciempiés* schools, but not for every child under 12 years old of age that walks along the treated census tracts.[7]

In an attempt to disentangle between these two, we estimate an specification at the school level, using data from school-issued accident alerts. This data contains any accident happening at the school premises or on the way to school. Figure 10 shows our event study estimates of the program's impact on accidents occurring on the way to school. We find null effects for the first two years post-treatment, and estimates that fluctuate between 0.4 and 0.1 thereafter, though these are not statistically different from zero.

We interpret these results as indicating a lack of improvement in the safety of children's daily commutes. This likely reflects the interaction of two opposing forces. On one hand, the program's road safety components—including the design of safe walking routes that minimize traffic exposure and educational activities promoting traffic awareness—should reduce the risk of traffic accidents. The presence of trained monitors who can stop traffic at dangerous intersections (as illustrated in Figure A.1) should also help in this regard. On the other hand, the program simultaneously reduces adult supervision relative to the counterfactual scenario where parents individually accompany their children to school. Under the *Ciempiés* model, just two trained monitors supervise groups of up to 50 children, creating a lower adult-to-child supervision ratio. This reduction in adult supervision may offset some of the program's safety benefits. In addition, if the mechanism that drives our estimated impacts on educational outcomes is increased attendance, we would expect more children on the streets during school commuting times, increasing the probability that an accident happens.

Our results demonstrate that the *Ciempiés* program successfully addresses the coordination challenges that prevent many families from maintaining consistent school attendance patterns. The program's most significant impacts are evident in measures of school retention and attachment; students are substantially less likely to switch schools or school branches. These effects are important, as dissatisfaction with public schools has increased post-pandemic, with many parents switching to substitutes of public schools (e.g., Francis and Goodman, 2025, in the U.S.). Our estimated effects grow stronger over time, suggesting that the benefits compound as families integrate the program into their routines.

While we do not find evidence of improved road safety for children, the program's primary value lies in its ability to reduce the logistical burden on families while maintaining

[7]The program may have also improved road safety for treated children, while creating other points of congestion/traffic accidents in nearby streets, offsetting the potential gains from designing safe paths to school with adult supervision.

adequate supervision of children's school commutes. The effects on educational continuity, achieved at a cost of just USD 285 per child per year, position *Ciempiés* as a highly cost-effective intervention for improving school attendance in urban settings.

These findings have important implications for education policy in similar contexts. The results suggest that relatively simple, community-based approaches to addressing transportation barriers can generate meaningful improvements in educational outcomes without requiring large-scale infrastructure investments or intensive individual interventions. The success of *Ciempiés* points to the potential value of addressing the practical, day-to-day challenges that families face in accessing education, rather than focusing exclusively on supply-side improvements or demand-side incentives.

5 Conclusion

This paper shows that relatively simple, community-based interventions can meaningfully improve educational outcomes by addressing the practical coordination challenges that families face when accessing education. The *Ciempiés* program in Bogotá successfully reduced school and branch switching by up to 6.8 and 10.9 percentage points, respectively, after two years in the program. These effects represent 41% and 42% reductions from pre-treatment rates in control schools, indicating that the program significantly increases children's attachment to their schools and reduces the disruptions that often force families to seek alternative schooling options.

While we find suggestive evidence of reductions in grade repetition and dropout, these effects are imprecisely estimated, likely reflecting our limited sample size for long-term impacts. Importantly, we do not find evidence that the program improved road safety for children, suggesting that the primary value of *Ciempiés* lies in addressing coordination and logistical barriers rather than direct safety improvements. At a cost of just USD 285 per child per year, the program is a highly cost-effective way to improve school retention compared to intensive, individualized interventions or formal transportation alternatives. These findings highlight the potential of place-based policies that improve educational opportunities within existing communities without requiring families to relocate or children to change schools, offering a promising model for addressing educational inequality in other urban contexts across LMICs.

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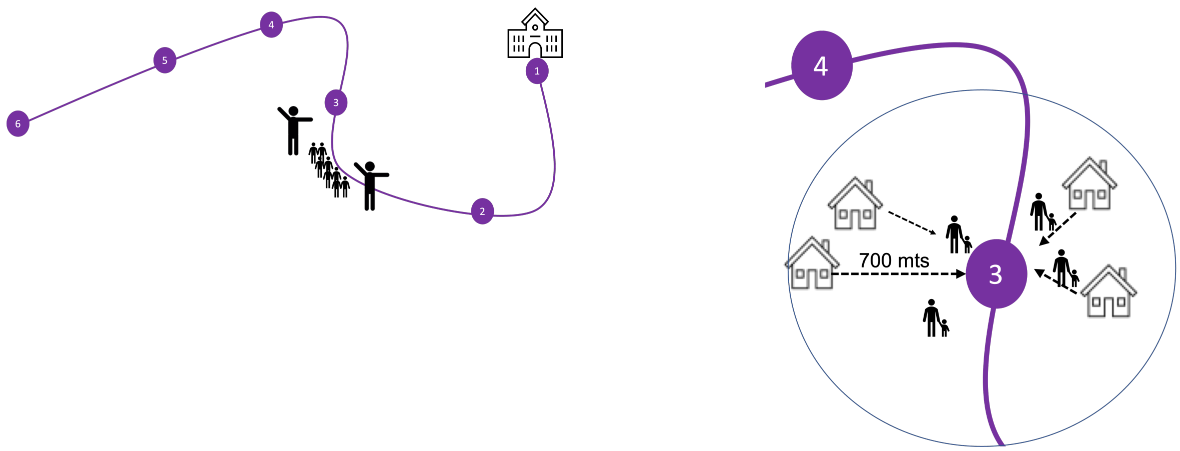
Figures and Tables

Figure 1: WSC Operation

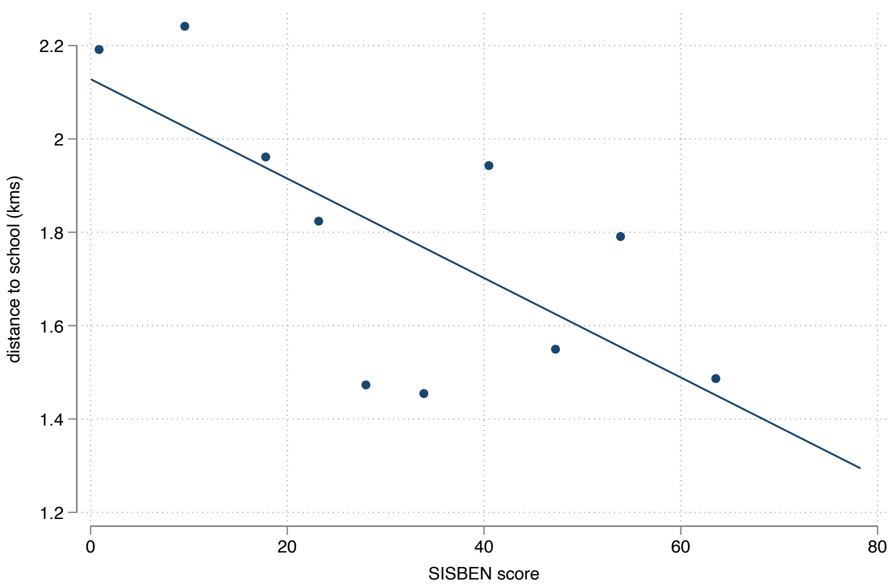
(a) Meeting Points

(b) Eligibility around Meeting Points



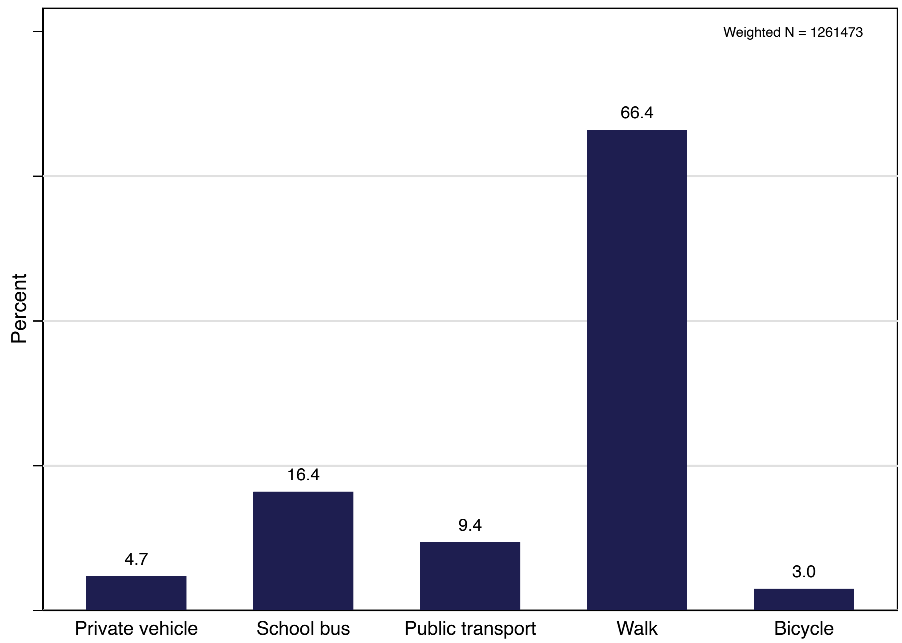
Notes: This figure shows a depiction of the WSC operation. Panel A shows the meeting points along the safe path to school. Panel B shows how a meeting point is used by the households living close by to drop and pick up children. Although there is an eligibility criterion on paper that says that students need to live within a 700m radius of any meeting point, this criterion is not enforced in practice.

Figure 2: Binscatter Regression of Distance to School on SISBEN Score



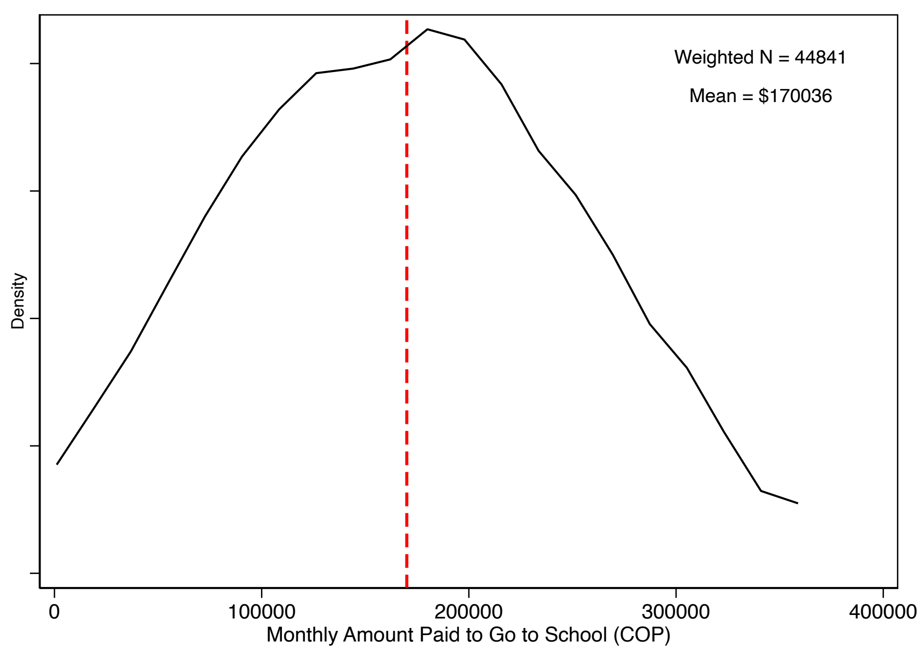
Notes: This figure shows the relationship between distance to school and SISBEN score, a household vulnerability index used to target social programs in Colombia. The sample corresponds to students in *Ciempiés* schools in 2021. The SISBEN score ranges from 0 to 100, with lower scores indicating higher vulnerability, and a score of 0 indicating extreme poverty.

Figure 3: School Transportation Modes in Urban Bogotá



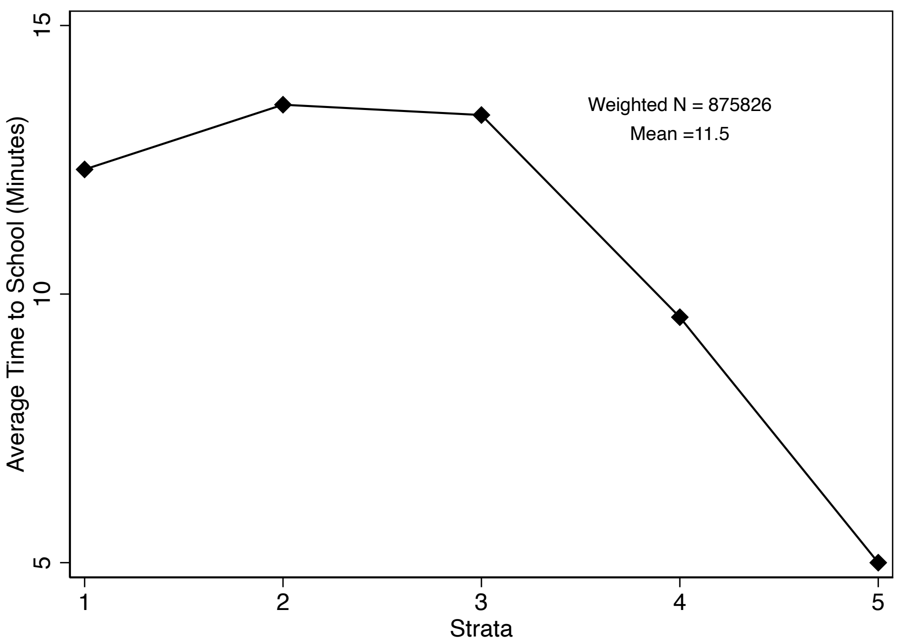
Notes: This figure shows the different transportation modes for children in public and private schools in Bogotá. The data corresponds to Colombia's 2024 QLS.

Figure 4: Commuting Costs to Schools



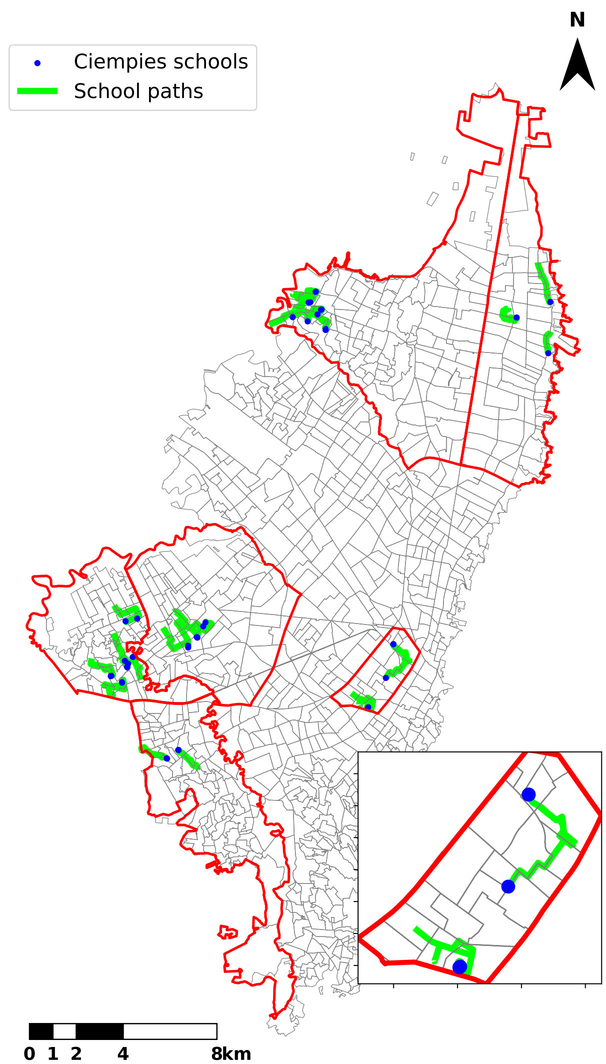
Notes: This figure shows the distribution of monthly commuting costs to school for the households that report paying for a transportation service. The dashed red line corresponds to the mean. The data corresponds to Colombia's 2024 QLS. All values are in COP.

Figure 5: Commuting Time to School by Socioeconomic Stratum



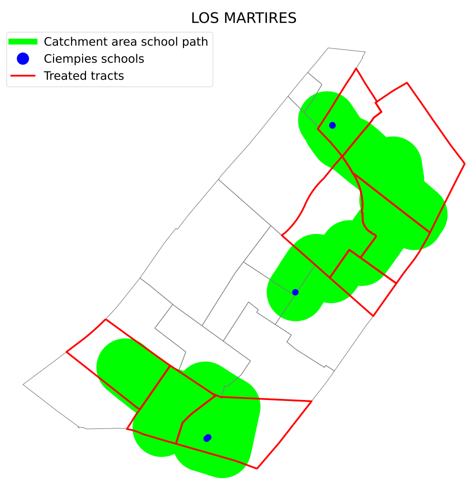
Notes: This figure shows the relationship between commuting time to school and socioeconomic stratum for children that commute to school by bike or on foot. The data corresponds to Colombia's 2024 QLS.

Figure 6: Ciempiés Schools and Paths in 2022



Notes: This figure shows the spatial distribution of *Ciempiés* schools and paths as 2022. As of that year, the program was only operating in 6 of the 19 *localidades* in Bogotá (circled) in red, but has expanded its operations to 9 as of 2024.

Figure 7: Defining Treatment and Control Census Tracts for Traffic Accident Analysis



Notes: This figure shows the procedure to define treated versus control census tracts using catchment areas around school paths. We define a treated census tract if its area intersects the catchment area of a school path in at least 50%.

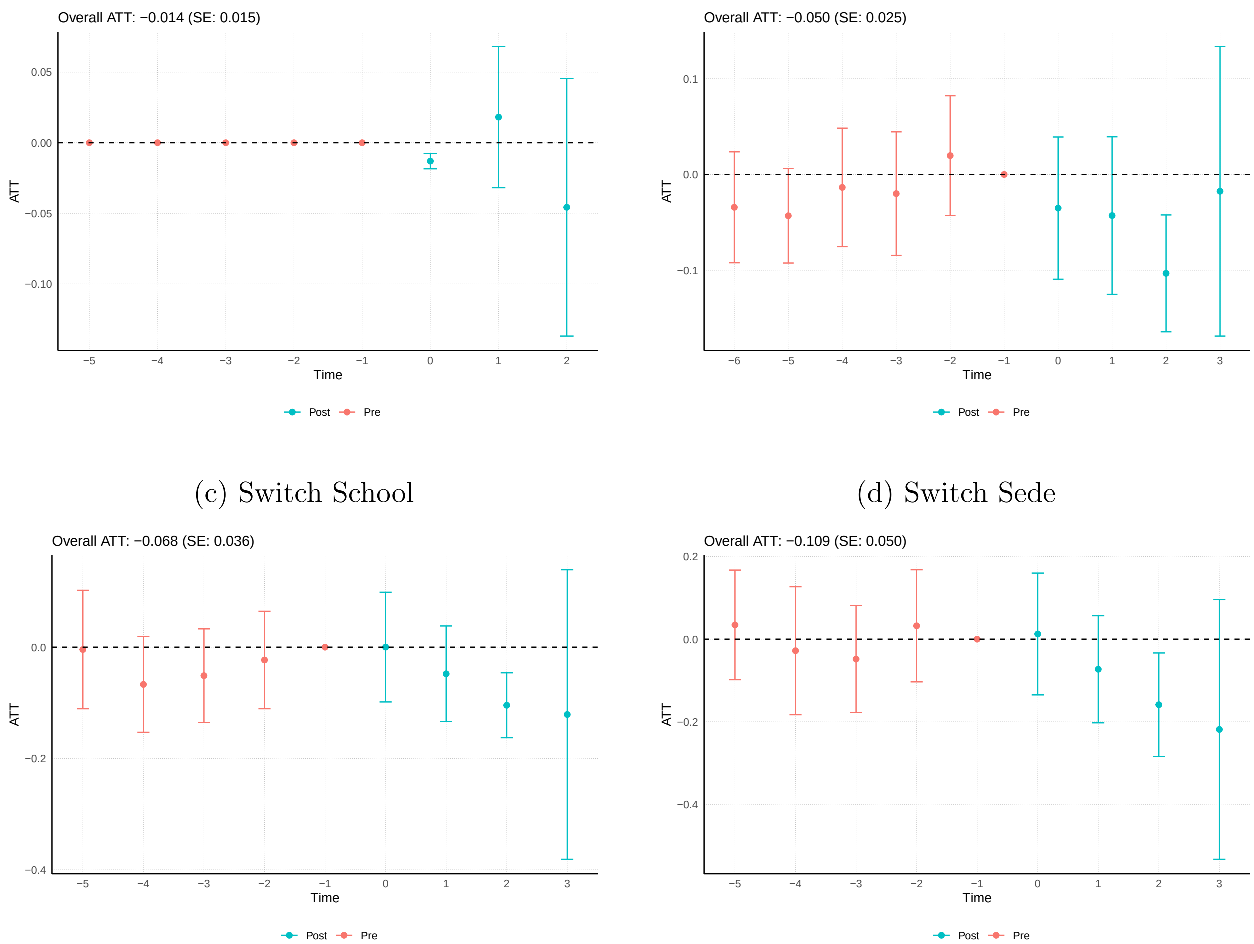
Figure 8: Impact of Ciempiés on School Trajectories of Children

(a) Dropout

(b) Grade Repetition

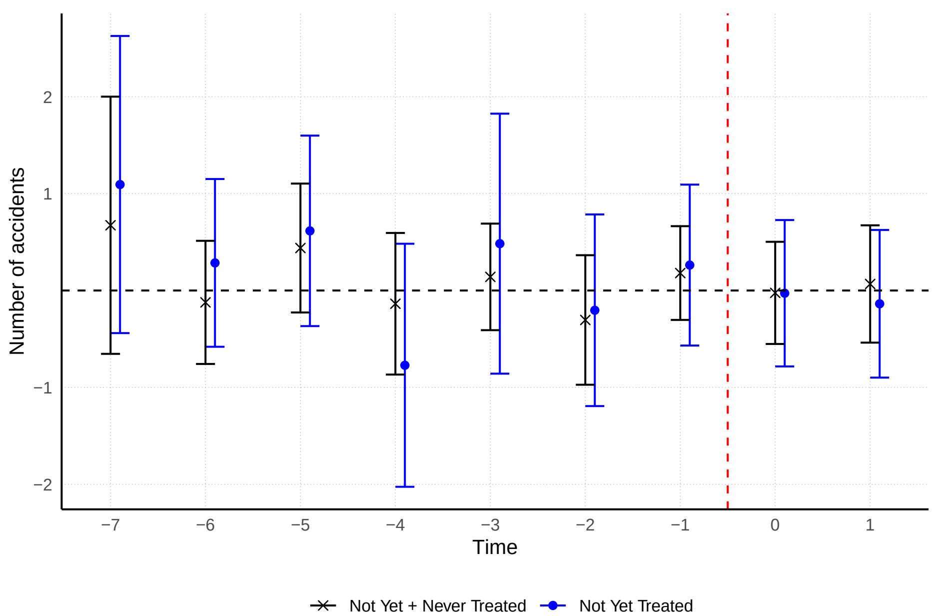
(c) Switch School

(d) Switch Sede



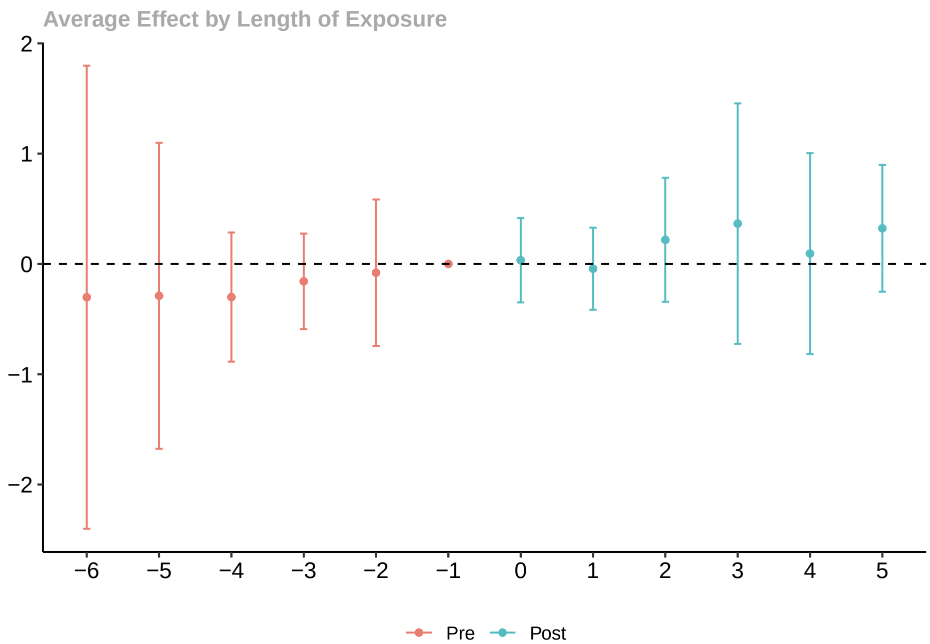
Notes: This figure shows the event-study aggregation of the estimated ATT for dropout, grade repetition, school switching and sede switching. The control group are the not yet treated students and the never treated students in school where the program is set to expand. 95% confidence intervals reported.

Figure 9: Results for Accidents Involving Children $\leq$ 12 Years Old



Notes: This figure shows the event-study aggregation of the estimated ATTs on census tract accidents involving children less than 12 years old. The estimates in blue only use not yet treated units in the control group, while the estimates in black use both not yet treated and never treated units. 95% confidence intervals reported.

Figure 10: Results for Road Accidents at the School Level



Notes: This figure shows the event-study aggregation of the estimated impacts of the program on school level accidents, relative to the year before the program. The data comes from the *Sistema de Alertas*. We compare earlier cohorts of entrants with not-yet-treated and never-treated schools. 95% confidence intervals reported.

Table 1: Descriptive Statistics for Ciempiés Children, 2021–2024

	2021	2022	2023	2024
Age	8.741	9.324	9.359	9.395
	(1.996)	(2.421)	(2.505)	(2.273)
Grade	3.276	3.778	3.775	3.785
	(1.839)	(2.272)	(2.336)	(2.089)
Share female	0.529	0.529	0.513	0.505
	(0.499)	(0.499)	(0.500)	(0.500)
Socioeconomic stratum	2.069	1.982	2.051	2.113
	(1.076)	(1.311)	(1.446)	(1.503)
Share victim conflict	0.035	0.042	0.090	0.077
	(0.184)	(0.200)	(0.286)	(0.267)
Share Venezuelan	0.095	0.122	0.150	0.133
	(0.293)	(0.327)	(0.358)	(0.339)
N	771	2139	2193	2820

Notes: This table shows means and standard deviations of student characteristics in our sample of Ciempies children, for the years 2021–2024.

Table 2: Descriptive Statistics for School Paths, 2020–2024

	2020	2021	2022	2023	2024
Length (km)	1.486	1.643	1.648	1.570	1.513
	(0.392)	(0.481)	(0.354)	(0.361)	(0.305)
Walking time (min)	38.286	41.389	41.500	41.890	40.915
	(9.736)	(8.558)	(8.767)	(7.176)	(6.452)
Share morning shift	0.571	0.481	0.475	0.439	0.434
	(0.514)	(0.509)	(0.506)	(0.502)	(0.500)
Share afternoon shift	0.214	0.407	0.425	0.439	0.434
	(0.426)	(0.501)	(0.501)	(0.502)	(0.500)
Share full shift	0.214	0.111	0.100	0.122	0.132
	(0.426)	(0.320)	(0.304)	(0.331)	(0.342)
N	14	27	40	41	53

Notes: This table shows means and standard deviations of school path characteristics, for the years 2020–2024. Length is measured in kilometers and walking time in minutes. Shift shares represent the proportion of paths in each shift (morning, afternoon, or ful-day).

Table 3: Covariate Balance Statistics

	Unweighted			Weighted		
variable	Non-Adopt	Adopt	Norm. Diff.	Non-Adopt	Adopt	Norm. Diff.
Gender	0.482	0.490	0.103	0.503	0.495	-0.080
Stratum	1.933	2.007	0.270	1.907	1.984	0.285
Number of accidents	0.313	0.158	-0.259	0.575	0.202	-0.511
Insufficient (math)	0.453	0.394	-0.516	0.443	0.392	-0.488
Minimum (math)	0.312	0.344	0.483	0.314	0.337	0.405
Satisfactory (math)	0.151	0.172	0.453	0.154	0.176	0.494
Advanced (math)	0.084	0.090	0.120	0.089	0.095	0.125
Insufficient (language)	0.119	0.108	-0.189	0.107	0.101	-0.120
Minimum (language)	0.441	0.421	-0.288	0.431	0.420	-0.165
Satisfactory (language)	0.321	0.339	0.256	0.331	0.337	0.107
Advanced (language)	0.118	0.131	0.235	0.132	0.142	0.186

Notes: This table shows how the 43 school branches that adopted Ciempiés between 2019 and 2024 compare to the 75 branches that did not adopt it by 2024. We present the means and standardized differences for each variable by adoption status, both unweighted and weighted by the number of primary school students in each branch. All variables are measured as shares, except for stratum, which is measured on a scale of 1–6, and number of accidents. All variables are measured in 2018 (the year before the first cohort adopted Ciempiés), with the exception of Saber variables, which are measured in 2017. This was the last time that standardized test scores were applied to 5th grade children in Colombia.

Table 4: Dynamic ATTs by Outcome

	Repetition	Dropout	Switch School	Switch Sede
ATT(-5)	−0.043	0.000	−0.004	0.035
	(0.019)		(0.038)	(0.048)
ATT(-4)	−0.014	0.000	−0.067	−0.028
	(0.024)		(0.031)	(0.056)
ATT(-3)	−0.020	0.000	−0.051	−0.048
	(0.025)		(0.030)	(0.047)
ATT(-2)	0.020	0.000	−0.023	0.032
	(0.025)		(0.031)	(0.049)
ATT(0)	−0.035	−0.013	0.000	0.013
	(0.029)	(0.003)	(0.035)	(0.053)
ATT(1)	−0.043	0.018	−0.048	−0.073
	(0.032)	(0.023)	(0.031)	(0.047)
ATT(2)	−0.103	−0.046	−0.104	−0.159
	(0.024)	(0.042)	(0.021)	(0.045)
ATT(3)	−0.018		−0.121	−0.219
	(0.060)		(0.094)	(0.113)
N	31 639	41 192	31 639	31 639
Time Periods	9	8	8	8
Cohorts	4	3	4	4

Notes: Standard errors in parentheses. ATT estimates from dynamic difference-in-differences model using the Callaway and Sant'Anna (2021) method. Negative (positive) numbers indicate periods before (after) treatment. Control group consists of never-treated + not-yet-treated units.

"Safe Paths to School: Evidence from Bogotá's Walking to School Caravans"

Online Appendix

Rafael Hernández-Pachón
Isabela Munevar

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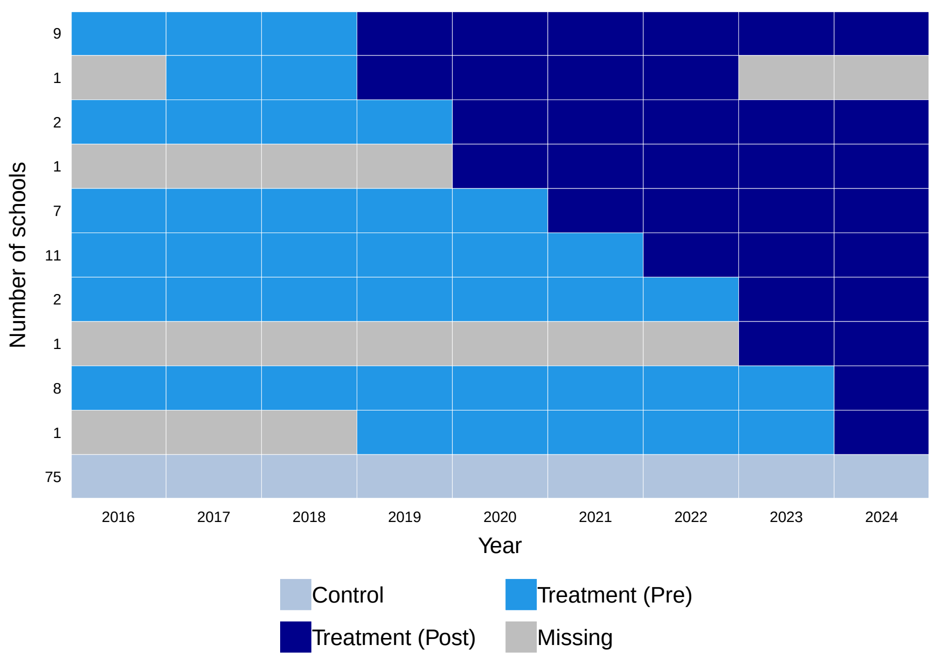
A Figures and Tables

Figure A.1: Example of Walking to School Caravan

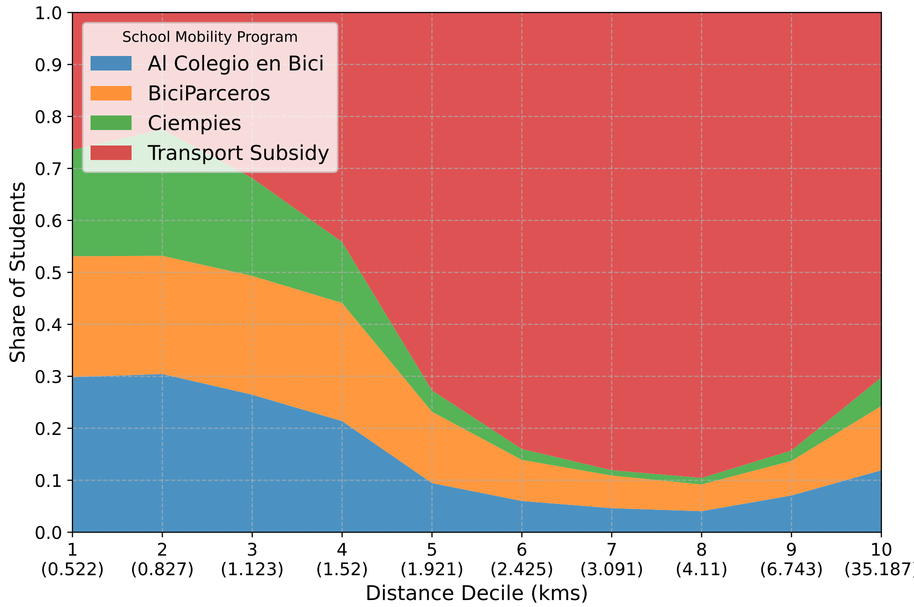
Notes: This figure shows photos from two Walking to School Caravans. The caravans are supervised by two adults, who ensure a safe commute to school. In the left panel we can see that these monitors often stop traffic for children to be able to cross the road.

Figure A.2: Treatment Rollout



Notes: This figure shows the treatment rollout from 2016 to 2024, marking the pre and post treatment periods for each cohort. The missing periods correspond to schools that opened or closed at some point during the sample period, and thus are not observed throughout the 9 years.

Figure A.3: Shares of Mobility Programs by Distance to School in 2024



Notes: This figure shows the shares of students in the different mobility programs by decile of distance to school. The max value of the distance to school in each decile is reported in parenthesis.

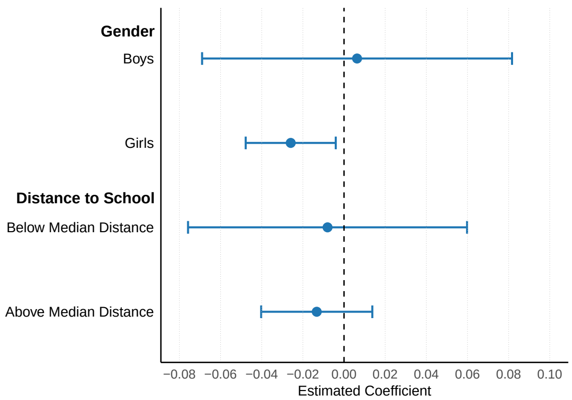
Table A.1: Description of Saber 5 Performance Levels

Level	**Description**
Advanced	Shows outstanding performance in the expected competencies for the evaluated subject area and grade.
Satisfactory	Demonstrates adequate performance in the required competencies for the subject area and grade. This is the level of performance that all or most students should reach.
Minimum	Shows minimum performance in the required competencies for the subject area and grade.
Insufficient	Does not surpass the most basic questions in the test.

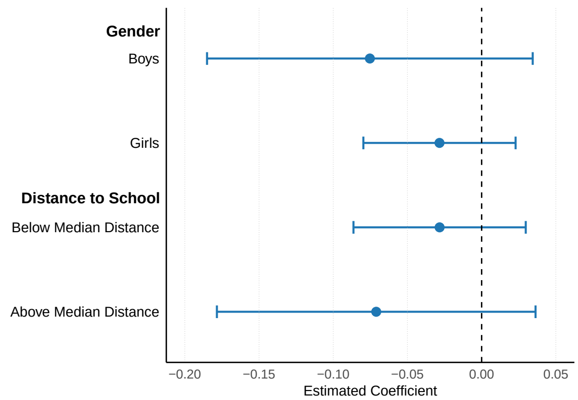
Notes: This table shows the performance levels corresponding to the Saber 5 assessment scale. Saber 5 is a standardized test administered to fifth-grade students in Colombia to evaluate competencies in math and language.

Figure A.4: Effects of Walking to School Caravans by Children Attributes

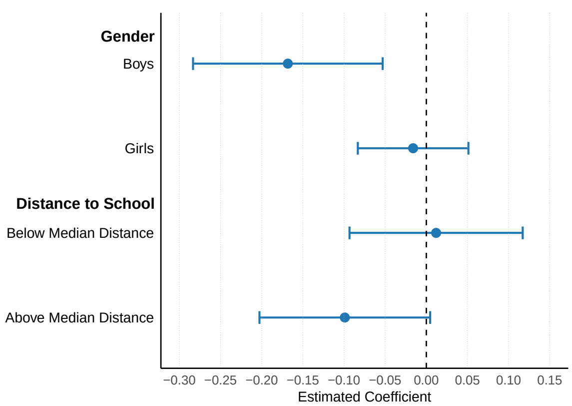
(a) Dropout



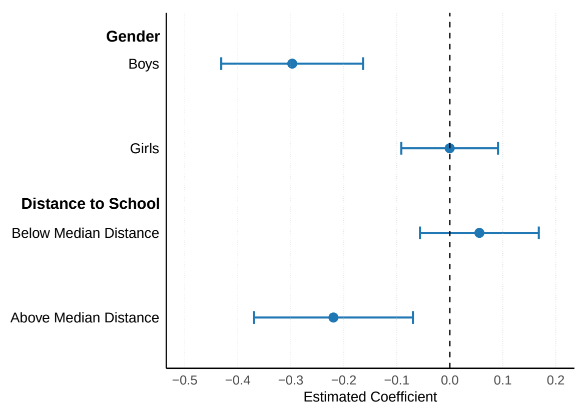
(b) Grade Repetition



(c) Switch School



(d) Switch Sede



Notes: This figure shows the overall ATT for the main four outcomes: dropout, grade repetition, school switching and sede switching, and different subsamples of children: boys, girls, living above median distance to school, and living below median distance to school. The control group is the same across all subsamples. 95% confidence intervals reported.

B List of Acronyms